

AI Usage Journal

Introduction

In this assignment a plethora of tools were used to accelerate the learning process and to utilize different mediums to convey the work that previously would have been reserved to specific domains. For instance with the use of AI it was possible to produce a fully functional web-based simulator with a professional looking visual appeal. This would not have been possible without a person with significant knowledge of front-end development. At the moment, there is a lot of changes in the way AI tools are being packaged into different tools. To make use of this a set of different tools were used to leverage the strength of each one while at the same being able to compare amongst them. In this assignment the following primary tools used are listed below.

NotebookLM

NotebookLM[1] which is provided by Google allows the user to upload a series of documents and other mediums and these are transformed into other medium such as mindmap, infographic and a slide deck. The biggest advantage of NotebookLM is that you can control the sources of information, so the outputted information is solely made from uploaded documents and no external sources are introduced unless a specific web features is used.

Perplexity

From my end I use Perplexity Pro[2] as a way of research. I have used it to find more information about the concept that I was looking for. One of the main advantages of Perplexity is that it offers clean source citations for each claim. This is very useful for technical research as it lets you to independently verify the claims from the original and verify that the conveyed information is not a result of hallucination. Another advantage of Perplexity is that it allows to switch between different models such as Claude Sonnet 4.6, ChatGPT5.4, and Gemini Pro 3.1, amongst others.

Claude

Another tool that was used is Anthropic's Claude[3]. Claude's subscription give access to multiple tools. There is the general Claude chat interface that was used for research, correction, and verification. It allows you to change amongst different models.

Apart from that there was also Claude Code that integrates with your IDE and provides and acts as an AI assistant. It is really helpful because it can handle a lot of boilerplate stuff that would be required to setup the environment required. Apart from that it essential for the coding of the simulator.

There is also Claude Design, which can produce high quality illustrations from concepts. You can give it some inspiration of styles, pass the content that you requires and it would arrange the content provided in a clean format to the desired style.

Another tools such as OpenAI ChatGPT and Google's Gemini were used but their use was minimal compared to other tools, more used in certain instances to compare the result obtained from one source to the other.

Ethical Considerations

In this section a list of ethical considerations related to the use of is being listed. One of them is data bias in LLMs, which can manifest in several ways. Since LLMs are trained on these models on data scraped from the public internet, the resulting outputs often lack guaranteed technical reliability. In software engineering contexts, LLMs can generate code based on hobbyist forums rather than professionally vetted, industry-standard sources, since they may be more common in the training data. Consequently, the models can output suboptimal design patterns that fail to meet rigorous engineering standards. Additionally, LLMs can display a strong temporal bias by favoring outdated libraries, frameworks, and coding practices. Legacy tools naturally dominate historical training data, prompting models to recommend them even after they have been deprecated due to security flaws, poor performance, or maintenance issues.

Adding to this, this can impact the evolution and adoption of programming languages. LLMs generate highly accurate code for established languages because those ecosystems offer extensive documentation and massive community representation. Conversely, newer or niche languages lack this volume of training data, resulting in poor model performance that actively hinders their adoption. This phenomenon of privileging dominant ecosystems while marginalizing newer alternatives is often referred to as The Matthew Effect [4] in academic literature.

In this specific assignment, data privacy concerns are not an issue as the research relies exclusively on publicly accessible academic sources. There are no datasets that contain personally identifiable or confidential information. Furthermore, because the work does not involve proprietary techniques, there is no risk of sensitive intellectual property or novel methodologies being inadvertently absorbed and exposed through the model's future training corpus. However, these are considerations that people using AI tools have to take into account.

The use of AI tools in this work was limited to supporting tasks such as literature discovery, language refinement, code support, and conceptual clarification. Nonetheless, as the author, all outputs were critically examined to ensure that concepts were correctly understood and that results were accurately interpreted. Any AI-generated text used to improve grammatical flow was reviewed and revised before inclusion in the document. This approach aligns with the principle that AI should augment, rather than replace, the researcher's intellectual contribution.

As an AI user, I am well aware of the risk of AI hallucination is well recognized and represents a material concern in academic work, where the integrity of cited sources is fundamental. While I have noticed that the reliability of contemporary models has improved incrementally over time, this limitation cannot be dismissed. Accordingly, AI was treated throughout this work as a suggester rather than an authority, therefore it is used as a tool to guide inquiry rather than to determine its conclusions. This principle was applied directly in the identification of papers mentioned in this work. The AI was prompted with the specific research topic and asked to suggest significant contributing works, along with a rationale for each paper's relevance. These suggestions served as an initial starting point, providing a high-level overview of the field and a preliminary literature map. However, every suggested reference was verified independently and each paper was located through established academic repositories, authorship and publication details were confirmed, and the claimed contribution was cross-checked against the paper's actual content. No reference was included in this work solely on the basis of an AI suggestion.

AI Integration Methodology

AI tools were integrated selectively into this work. The actual process of learning, understanding and analysis was retained. It was just included into tasks that would not directly contribute to the main objective of this assignment to learn the concept of “Non-Maximum Suppression and Average Precision”.

Tasks Where AI Was Used

Literature Discovery

AI was used to identify key contributing papers in the machine vision and NMS domain, providing an initial map of the field. All suggested references were subsequently verified independently through academic repositories before inclusion.

Code support and debugging

AI assisted in drafting and reviewing Python implementation code, particularly for understanding library-specific syntax and resolving integration issues. All generated code was critically reviewed, tested, and validated against expected outputs before adoption.

Language refinement

AI was used to improve the grammatical flow and academic register of written sentences. During the write, there are instances that you feel that the flow of the text provided is not natural and requires some modifications to sound more polished. The outputted text is often re-tweaked as sometimes the original meaning that I want to convey would be changed. Sometimes it is also used to find a more adequate word given the context of the text.

Conceptual clarification

Where unfamiliar concepts arose, AI was used to explain to me the concept in a familiar way. Therefore, given that my background is in electrical engineering, I would often use the following statement

Explain to me _____ as an electrical engineer with no background in machine learning.

Given that I have memory on, it would take context from previous questions that I have asked and use that as an example. For example, I have recently asked about alarms in SCADA system and the model gave the following output.

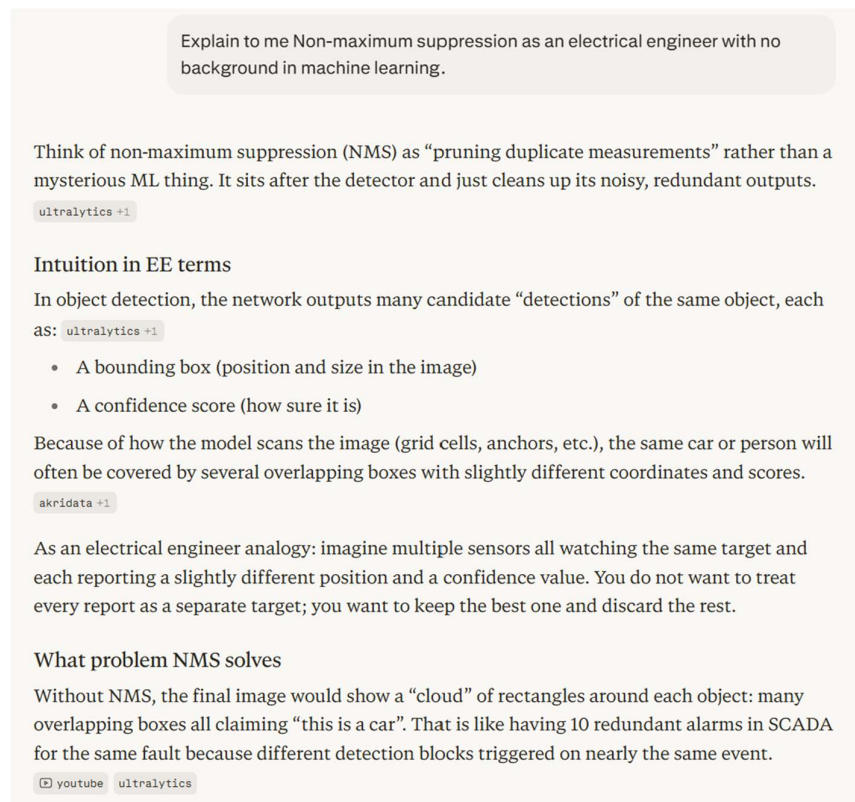


Figure 1: Snapshot from Perplexity Pro using “Learn step-by-step” tool being used.

Tasks Deliberately Avoided

Algorithm design

In the case of the Jupyter notebook, the AI was used as help in the syntax rather than in the logic. Therefore, for example if I need to sort a table, the AI would help assist in the sorting part, but before I would manually check the input list and the output list and confirm that the output makes sense.

Quantitative result interpretation

The metrics, detection outputs, and comparative evaluations were analysed and interpreted by the author. Apart from completely missing the point of the assignment and get the academic background that I am looking for when I have signed up for this MSc, delegating this to AI would risk misrepresentation of results, especially in edge cases.

Drawing conclusions

No AI tool was used to formulate findings or recommendations. Conclusions reflect my opinion after understanding the topic.

This selective approach reflects the view that AI is most valuable as an assistant. In this specific task, given that this is an assignment intended for academic I need to get down to the nitty-gritty to really understand what I am doing. Let’s for the purpose of this conversation assume that this was not the case. Let’s assume that this was a work-related task where the main objective in that scenario would be to get the task done. Hallucinations are still a present issue, so if I am presenting a piece of work that would be

under my responsibility, I have to make sure that I have independently verified. The AI can do the work, but ultimately, the verification of that work would still be withing my responsibility.

Prompts and Responses

NotebookLM

During the research process, NotebookLM was used.

For example in the case of the bounding boxes, the following was developed.

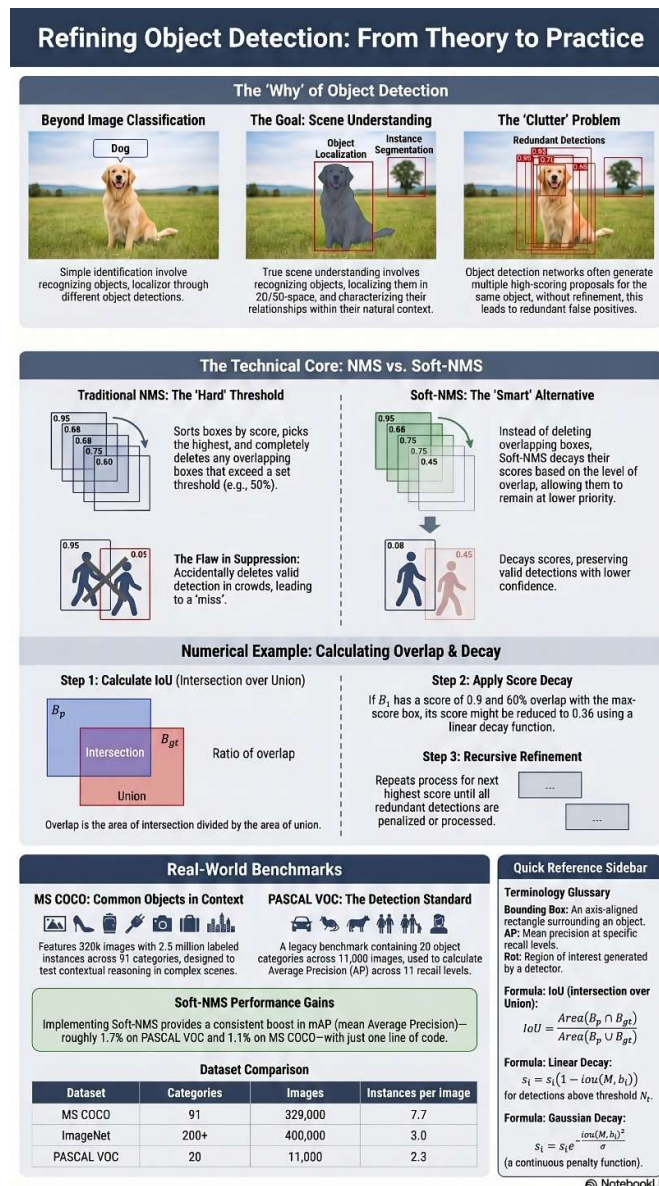


Figure 2: Output from NotebookLM

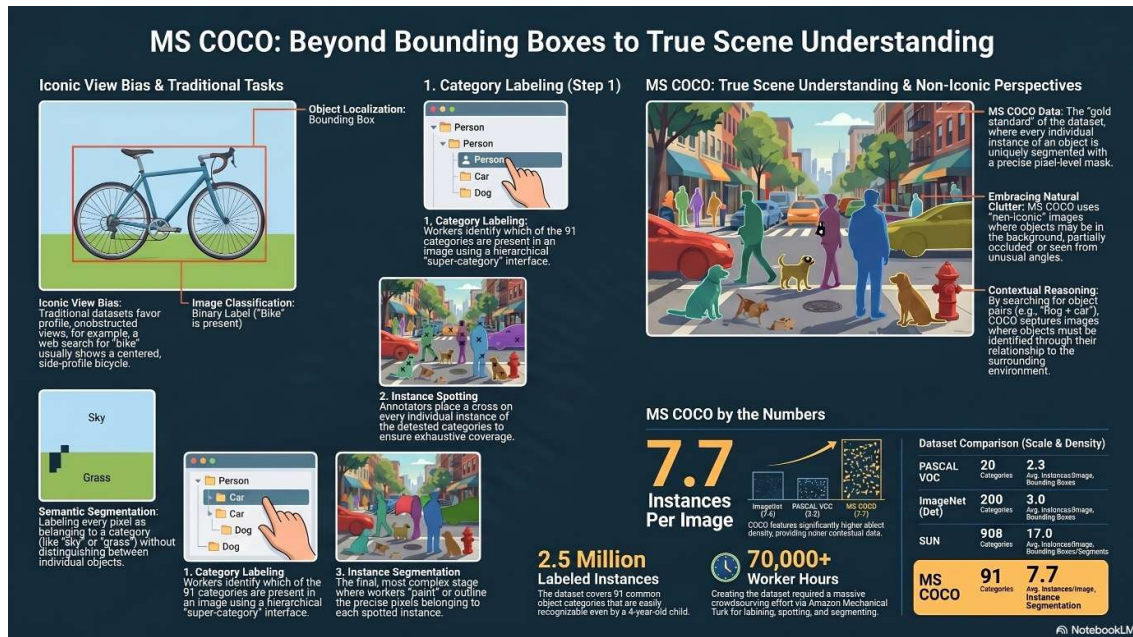


Figure 3: Output for COCO

Simulator

As previously described AI was used for multiple tasks. In this section I am going to specifically the describe the process taken to generate the simulator.

The first step after understanding the concept of NMS, a round sketch was done. Eventually, after understanding more the concept. A more accurate sketch was drawn still by hand at this process as shown in the image below. During the sketching of this image, I had my laptop and was jotting down rough notes of what different elements of the design shall do.

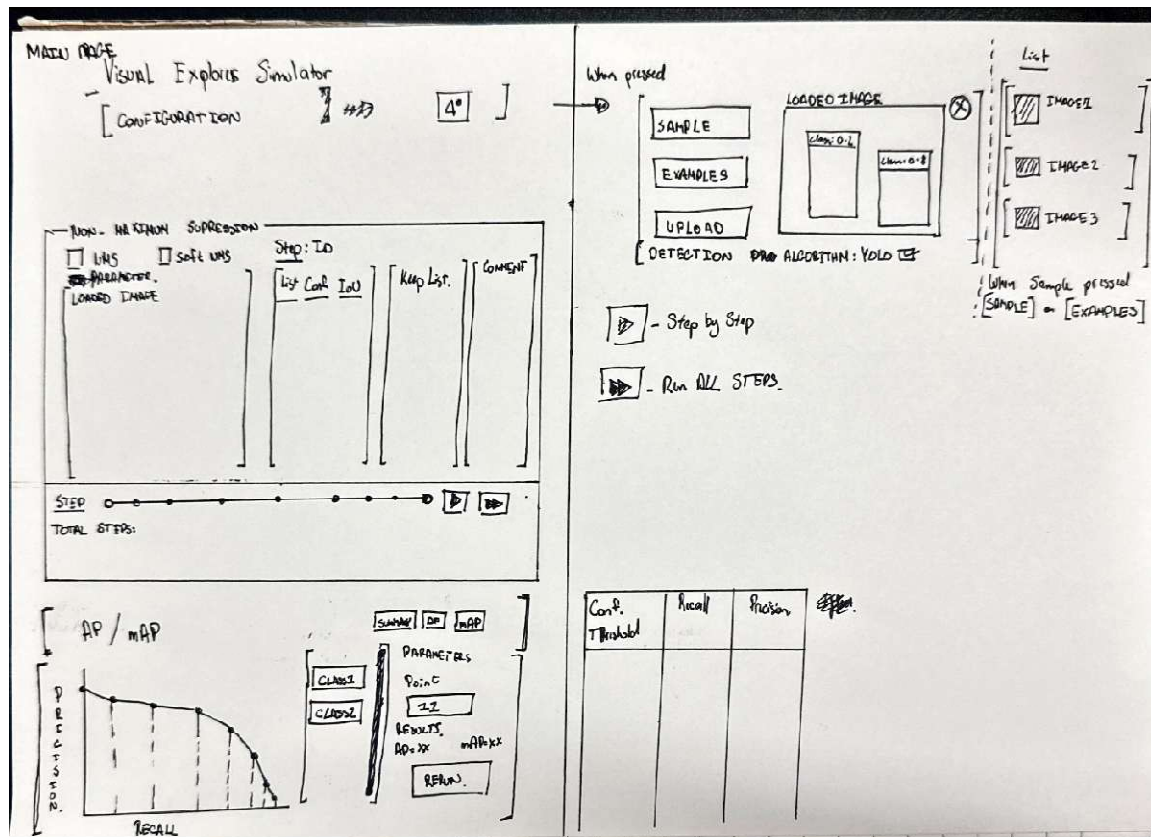


Figure 4: Hand drawn sketch

The full list of requirements.txt that was written by hand, can be found at the back of the documents.

Following these, I have prompted Claude to write a prompt that I shall give to Claude given that I have this sketch and this list of requirements.

This was the resulting output:

I am building a simulator and I have two inputs for you: 1. A rough requirements document (attached) 2. A hand-drawn sketch of the GUI (attached image) Please do the following in order: --- STEP 1 — UNDERSTAND & CLARIFY Read both inputs carefully. List any ambiguities, contradictions, missing information, or unclear interactions you find. Ask me to resolve them before proceeding. --- STEP 2 — GUI DESCRIPTION Once clarifications are done, describe the GUI sketch in structured text: identify every panel, widget, button, display area, and layout region. Give each a clear name. Describe interactions where visible. --- STEP 3 — PRODUCE spec.md Write a clean, structured Markdown specification with these sections: - Project Overview (1 short paragraph) - Simulator States (e.g. idle, running, faulted — list all) - UI Components (each with: name, location, purpose, behavior) - Data Flow (what inputs drive what outputs) - Acceptance Criteria (how to verify each component works) - Constraints (tech stack, platform, known limitations) - MUST / SHOULD / MAY requirements table --- STEP 4 — PRODUCE CLAUDE.md Write a CLAUDE.md file for Claude Code with: - Project purpose (2 sentences max) - Tech stack and file structure - Reference to spec.md for full requirements - Key conventions Claude Code

must follow - Any explicit things Claude Code must NOT do --- Do not proceed from one step to the next without my approval. Start with STEP 1 only.

The resulting output of this section, was a list of 32 questions addressing an ambiguities. The full list of questions and their answers can be found in the appendix.

Following this list, another list of questions was given as shown below.

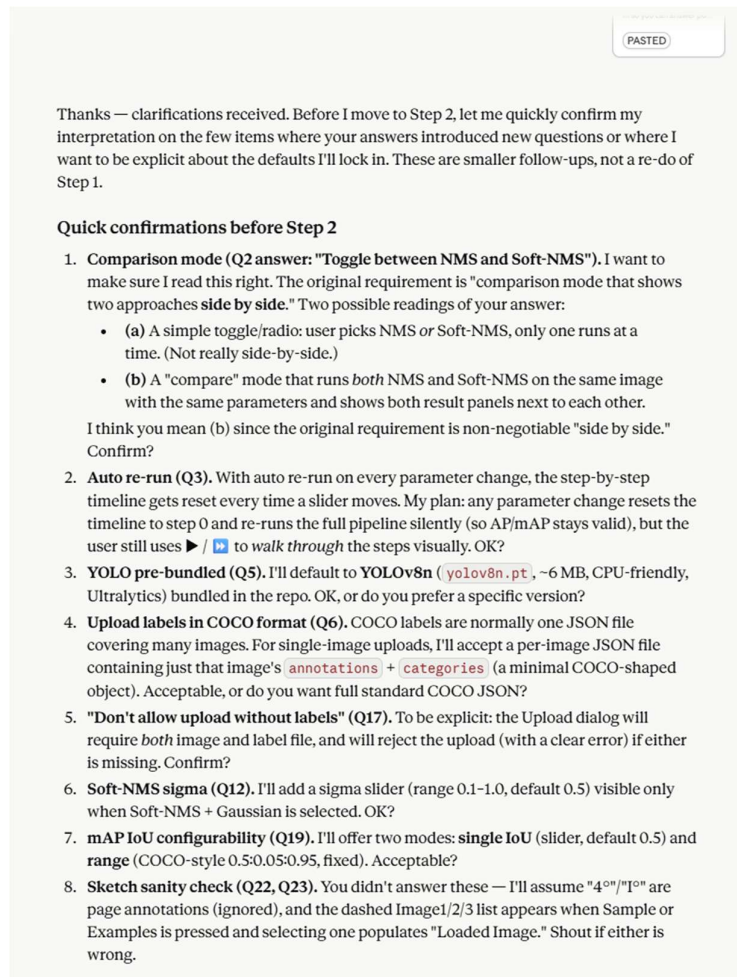


Figure 5: List of 2nd Batch of Questions from Anthropic Claude

Eventually after this step a list of specifications defining the GUI given which I have accepted. This section is shown in the image below.

STEP 2 — GUI Description

Below is a structured description of the GUI as drawn in the sketch, refined by your clarifications. Names in `code style` are the canonical identifiers I'll use throughout `spec.md` so labels stay consistent across docs and code.

Page-Level Layout

The application is a **single-page interface** titled "Visual Explorer Simulator". It is organized as a vertical stack of three primary sections, top to bottom:

1. `ConfiguratorPanel` (top)
2. `NMSPanel` (middle)
3. `APmAPPPanel` (bottom)

The page header sits above all three sections and contains only the title text. Background is light-themed.

Section 1 — `ConfiguratorPanel`

Location: top of page, full width. **Purpose:** Select the image (and labels) that will feed the rest of the simulator.

Sub-components

- `ConfiguratorHeader` — contains the section title "Configuration" and a **collapse toggle** (`ConfiguratorCollapseButton`, the small "X" / chevron in the sketch). When collapsed, the entire panel shrinks to a thin bar showing only the title and a `LoadedImageThumbnail` preview icon of the currently selected image. The user **must expand** the panel to change the image — interactions are disabled while collapsed.
- `SourceButtonRow` — three mutually exclusive buttons:
 - `SampleButton` → on press, reveals `ImagePreviewStrip` populated from the bundled COCO subset.
 - `ExamplesButton` → on press, reveals `ImagePreviewStrip` populated from the curated examples list (driven by `examples.txt`), with each preview showing its associated description tooltip.
 - `UploadButton` → on press, reveals `UploadDialog` containing two file inputs: `ImageUploadField` and `LabelUploadField`. Both are required; submission is rejected with an inline error if either is missing or the label JSON is malformed. Successful uploads are written to a persistent uploads folder and appear in the Sample list on subseq ↓ sessions.
- `ImagePreviewStrip` — a horizontal row of thumbnails (`ImagePreviewTile` × N)

Figure 6: Output from Anthropic Claude

Eventually, following this section, `spec.md` and `claude.md` were generated as shown in the image below. The full list of `spec.md` and `claude.md` can be found in the Appendix.

STEP 3 — `spec.md`

Created a file, read a file >

`spec.md` is ready. It covers Project Overview, Simulator States (with a transition table), every UI Component grouped by panel, a Data Flow diagram showing what feeds what, Acceptance Criteria as a verification table, Constraints (tech, platform, limitations), and a MUST/SHOULD/MAY priorities table.

A few decisions I locked in that you should double-check before I move to Step 4:

- **Backend framework:** FastAPI as the recommended default, with Flask called out as acceptable. If you have a strong preference, say so and I'll fix it in both files.
- **Frontend:** vanilla JS with Alpine.js or htmx allowed as MAY. No heavy SPA framework.
- **Plotting:** server-side Matplotlib *or* client-side Chart.js — left as MAY so whoever implements can pick.
- **Compare mode parameter sharing:** documented as a v1 limitation that both sides share parameters. If you want independent parameters per side, that's a real change.

Let me know if any of those need adjusting, otherwise say the word and I'll proceed to **Step 4** — `CLAUDE.md`.

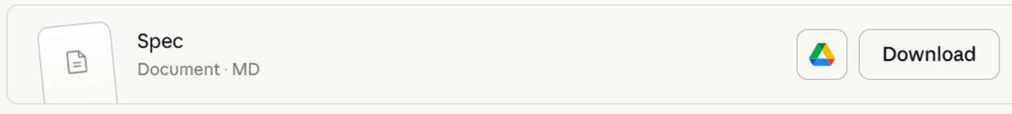


Figure 7: Output of `spec.md` as generated by Anthropic Claude

Moreover, the following `claude.md`, was generated.

STEP 4 — `CLAUDE.md`

Created a file, read a file >

`CLAUDE.md` is ready alongside `spec.md`. Both files are above for download.

A quick map of what's in `CLAUDE.md`:

- **Project Purpose** — two sentences, as requested.
- **Tech Stack** — locked-in choices (FastAPI, vanilla JS / Alpine / htmx, Chart.js or Matplotlib).
- **File Structure** — proposed layout that cleanly separates `core/` (pure algorithms) from `api/` (FastAPI) so the algorithms stay testable and reusable.
- **Reference Documentation** — explicit "spec.md is the source of truth" rule, plus a directive that the canonical component names from `spec.md` flow into code identifiers, DOM, and API payloads.
- **Key Conventions** — code style, the pure-`core/` architecture rule, the `Step` trace contract that the frontend renders from, state management, error/warning handling, testing minimums, performance notes, and commit hygiene.
- **What Claude Code Must NOT Do** — a long, deliberate list. Highlights:
 - No GPU code path, no runtime downloads, no external API calls.
 - No frontend framework with a build step (rules out React/Vue/Vite/etc.).
 - No database; filesystem persistence only.
 - No re-implementing algorithms in the frontend.
 - No accepting uploads without labels; no label formats other than per-image COCO JSON.
 - No silently resolving `spec.md` ↔ `CLAUDE.md` conflicts — flag them.

Two things worth your attention before you hand this off:

1. **Plotting choice is still "pick one."** I left Chart.js or server-side Matplotlib as a one-time decision the implementer makes. If you want me to lock one in (Chart.js is lighter; Matplotlib avoids a JS dependency), say which.
2. **The `Step` trace contract in `CLAUDE.md` is mine** — I introduced it because the spec's per-step UI requirements need a concrete data shape to drive them. If you want it moved into `spec.md` instead (so it's a binding interface rather than an implementation convention), I can do that.

All four steps are done. Let me know if you want adjustments, or if you're good to hand these to Claude Code.

Figure 8: : Output of `claude.md` as generated by Anthropic Claude

Eventually the following were used and imported into Claude Code. Initial iterations had issues with YOLO, however these eventually were corrected over time.

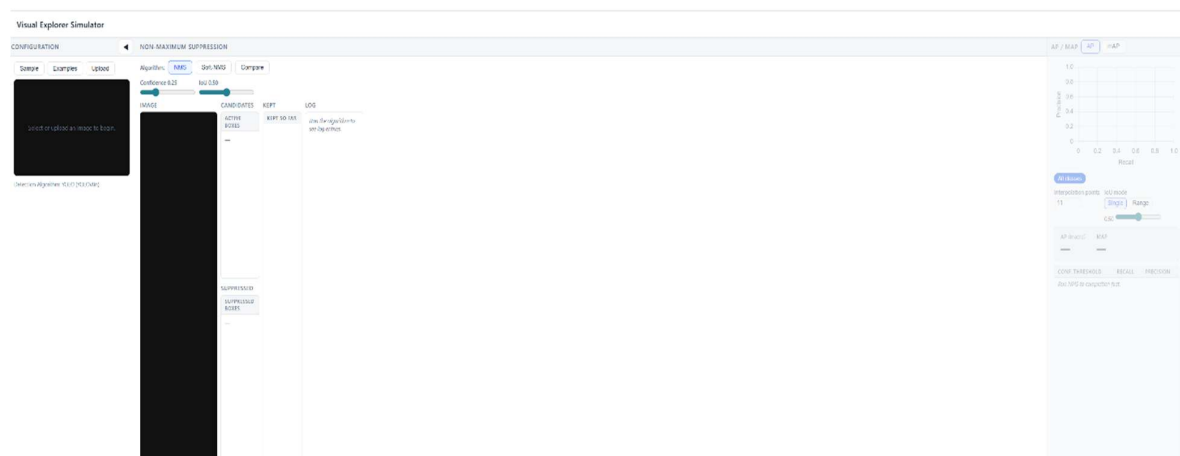


Figure 9: Early Iteration

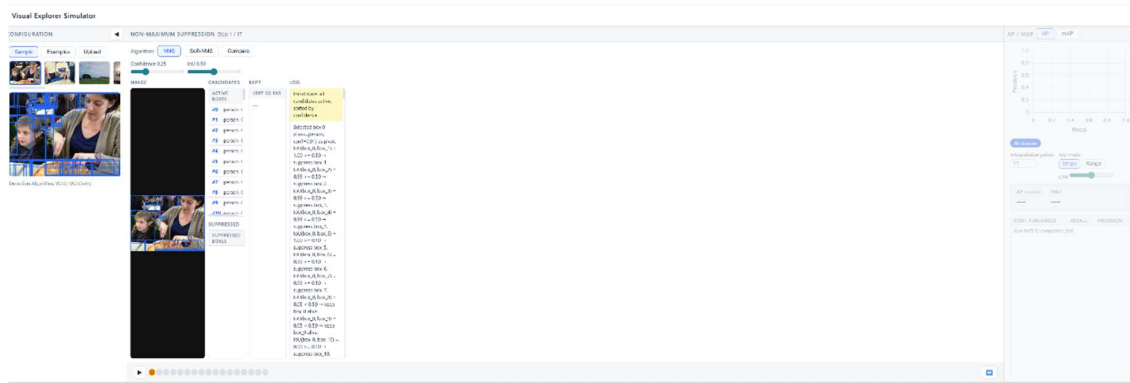


Figure 10: Progressed iterations

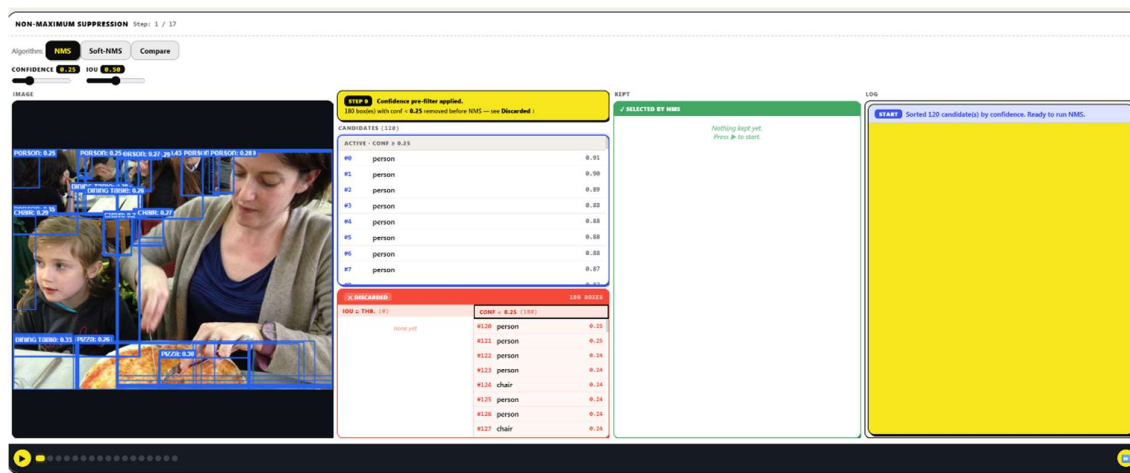


Figure 11: Final Iteration

Improvements and Contributions.

In this assignment, AI tools contributed meaningfully across several deliverables in this project, with the most substantial impact felt in the simulator development, literature discovery, and language quality of written outputs.

Simulator Development

One of the biggest contributions of AI was in building the interactive NMS simulator. Without AI assistance, specifically in this case Claude Code, producing a fully functional web-based simulator with a Flask/FastAPI backend and a polished HTML/CSS frontend would have required significant front-end development expertise that falls outside my background as an electrical engineer. To do so it would have required extensive more time to familiarise myself with languages since I am not fluent with them and use them on occasions.

There were instances when the code especially the part when it comes to get the pre-NMS outputs was not working well. To solve it there were multiple instances where Claude Code was prompted and the resulting output was shared again with Claude until it was resolved. Although this steps looks like a waste of time, however in reality to understand the code structure and fix it manually would most probably take much more time.

Adversarial Quiz

For the Adversarial Quiz, the AI tools contributed to the suggest plausible distractors and give ideas that I may have not thought about. The idea to come up with a misconception just from my own understanding would be biased towards, however like this I managed to capture a greater range of questions. Eventually it was used to help summarize the questions into the required 2 page format.

Conceptual Clarification

AI was also used as a learning tool to explain information. In the past, when I had to learn something that I was unfamiliar with, I had to do research blog posts, find Youtube videos or possibly manage to find a book during self-learning. My biggest issue was that I would have questions that would basically determine if what I thought that I have learned or no. In the past I either had to find a lecturer to answer or else ask on a forum. I know that in a forum the source may not be reliable but if the answer is backed up with a meaningful rationale, you can critically determine if it makes sense or no.

When unfamiliar concepts arose, Perplexity's "Learn step-by-step" feature was used with a deliberate framing strategy: prompting the model to *explain a concept to an electrical engineer with no machine learning background*. Because Perplexity Pro retains memory across sessions, it contextualised explanations using prior domain-relevant examples, such as SCADA alarm systems, making abstract computer vision concepts more intuitive and directly informing the analogies used in the study notes. I know that LLMs can hallucinate and I am well aware of that. So what I did when an issue like this arose, I would just still use the same approach that I did for when I used the forum to check the reason.

Reflection

After completing this project, I would say that AI has definitely played a crucial part in helping. Lets start with the research part. I could have been biased, but this is not something exclusive to AI as could also happen in a conventional approach where I start doing a research on something, find a piece of literature that takes you to a certain direction and automatically, the continued literature would on particular path. In general, I found that AI would be better to give you a general perspective and show you avenues that normally you would miss.

Apart from that, as I already said AI tools give me a new medium to how I can convey information. Before the proliferation of AI tools, the common medium would have been writing. Now the medium of tools is very much diverse. For instance, before I could have conveyed the concept of NMS through writing. To convey another medium I would have had to learn a completely new subject to produce a simulator. Now only using the medium that I already know, which is that of writing I was able to convey new content. I am not saying that the process is frictionless. Early simulator iterations failed due to YOLO integration issues, and several debugging cycles were needed before a working build emerged. Claude occasionally changed the intended meaning when refining sentences, requiring careful re-reading and manual correction every time. At this stage, I still believe that having absolutely zero knowledge on computer programming/ system would still be difficult to produce such results but has lowered the barrier of entry by a lot.

The summarize my observation is that AI shifted my effort rather than eliminated it, however, there was less time wasted on boilerplate stuff and more time on understanding the concepts. The risk of uncritically accepting output was real, particularly with code logic and cited references, and staying accountable required constant discipline. AI genuinely saved time in two areas: bootstrapping the simulator and mapping the literature. What would have taken days of front-end trial and error was compressed into structured iterations with Claude Code. Similarly, having a preliminary paper list to verify against was far faster than starting blind. However, AI was not frictionless. Early simulator iterations failed due to YOLO integration issues, and several debugging cycles were needed before a working build emerged. Claude occasionally changed the intended meaning when refining sentences, requiring careful re-reading and manual correction every time. The most honest observation is that AI shifted my effort rather than eliminated it. The idea is that less time on boilerplate, more time on verification.

Reference

Tools used

Perplexity Pro was used.

(Multiple models Anthropic Claude. Sonnet 4.6 with thinking, ChatGPT). I have used multiple tools as well like step-by-step.

Google NotebookLM

Anthropic Claude

- Chat for Claude using Sonnet 4.6 (For light questions) , Opus 4.7 and Opus4.7 with Adaptive Thinking.
- Claude Code was used within the VSCode IDE.
- Claude Design

ChatGPT

ChatGPT for small tasks like image generation.

Bibliography

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- [4] F. Gu, Z. Liang, H. LI, and J. MA, 'The Matthew Effect of AI Programming Assistants: A Hidden Bias in Software Evolution', Oct. 13, 2025, *arXiv*: arXiv:2509.23261. doi: 10.48550/arXiv.2509.23261.

APPENDIX

Requirements.txt

This document contains the initial wrote written down during when the final hand-drawn sketch was being drawn.

Non-Maximum Suppression and Evaluation Metrics

I am building a simulator to do showcase different concepts such as Non-Maximum Suppression and Evaluation Metrics, NMS algorithm, IoU threshold, Soft-NMS, precision-recall, Average Precision, mAP.

The Simulator is to be built with a backend in Python and a front end in HTML/CSS such that it can be hosted as a web interface. A functional, interactive tool built in Python (Streamlit or Gradio) or a web-based framework.

- Users must be able to adjust parameters (e.g. thresholds, kernel sizes, confidence levels) and see the visual output update in real time.
- Include a comparison mode that shows two approaches or parameter settings side by side.
- The simulator must run without a GPU. CPU-only execution is a hard requirement.
- Code must be clean, commented, and executable by your peers with no additional setup beyond the provided requirements file.

Structure

The simulator is split in 3 parts.

The top part is the Configurator.

Middle part is the NMS Section

Third part is the AP/mAP Section

Configurator

The configurator

The configurator is used to select the image to be used. Inside there shall be the following buttons:

Sample: When pressed opens a list of images that are loaded from a predetermined dataset. For example for this case the COCO can be used.

Examples: It contains a list of images from the dataset that are noteworthy. For example there is a boundary condition. When pressed a list shall opened with these selected images and a description why the image is example worthy shall be given. To select this list during development a txt file shall be in the development package to select which images and a short description is give why it is example worthy. During development, put placeholder text.

When Upload is pressed, it shall open two section to upload, one for image and the other one for uploading the Label. The uploaded images and labels shall be saved in a specific folder and added to the Sample List.

In all 3 cases, once the image is loaded, it shall be shown in 'Loaded Image' and shown including the bounding boxes.

At the bottom show the model used for creating the bounding boxes. At the moment just YOLO.

The configurator can be closed and the loaded image is previewed as a small icon on the side of the button.

NMS

The second step is the NMS Section

This is the most important step of the simulator.

On top it shall show which algorithm to choose from NMS, Soft-NMS.

It shall also let you chose the Confidence Threshold to select and the IoU Threshold.

Below it shall showcase the loaded image with the Number of bounding boxes at each step. Therefore for example in the first step, it shall show all the bounding boxes generated from YOLO. In the second step, the loaded image box shall not show the removed bounding boxes.

Next to it shall show a list of the 'Bounding boxes list' with the detected and next to it the 'Keep list' for NMS. After each Step, the removed ones shall be removed from the 'Bounding boxes list'. Next to it shall show the Comment section, showing the logic taking place. Like IoU calculations, Threshold Compare etc for NMS. Each boxes in question shall be highlighted in the loaded image if pressed in the comment section.

In the bottom shall show the timeline of all the step that have taken place. Eventually showing count until the full has been reached. It shall show a timeline which the user can browse through it.

###AP/ mAP

Greyout until all steps in NMS run.

When all run, it shall show Average Precision and mean Average Precision. On the side shall show a list for each class to calculate for each Precision/Recall graph.

The user shall be able to set the number of points for calculation and then a summary of the results.

There shall be a button to so a recall.

STEP 1 — Ambiguities, Contradictions & Missing Information

List of ambiguities, contradictions & missing information as asked by Claude.

STEP 1 — Ambiguities, Contradictions & Missing Information Below are the issues I found, grouped by section. I've numbered them so you can answer point-by-point.

A. General / Tech Stack 1. Framework choice. The requirements say "Streamlit or Gradio or a web-based framework" and also "backend in Python and a front end in HTML/CSS." These point in different directions. Streamlit/Gradio don't give you direct HTML/CSS control; a Flask/FastAPI + JS frontend does. Which do you want? The sketch (with its custom panels, timeline scrubber, highlightable boxes) looks closer to a custom Flask/FastAPI + JS build than to vanilla Streamlit. - Use Flask/FastAPI + JS frontend does. 2. "Comparison mode" (side-by-side). The requirements explicitly mandate this, but I see no comparison panel in the sketch. Where should it live? Inside the NMS section (two NMS configs side by side)? Or a separate toggle for the whole simulator? Toggle between NMS and Soft-NMS 3. Real-time update expectation. "See visual output update in real time" - does this mean every slider drag re-runs NMS instantly, or is there a "Run" button? The sketch shows step-by-step + run-all buttons, suggesting manual triggering. Re run automatically

B. Configurator Section 4. Sample dataset. "COCO can be used" - do you want me to bundle a small subset of COCO images locally (since no GPU and no guarantee of internet at runtime), or pull from the COCO API? How many sample images? Save the photos locally, small subset 5. YOLO model. Which YOLO version/size? (YOLOv5n, YOLOv8n, YOLO11n...) Smallest CPU-friendly variant by default? And should the model be downloaded at first run or pre-bundled? Pre-bundled 6. Label format for uploads. What format do you expect users to upload labels in? COCO JSON? YOLO txt (class cx cy w h)? Pascal VOC XML? This matters for parsing. Same as COCO 7. "Examples" file format. You mentioned a txt file listing noteworthy images + descriptions. Should this be a simple examples.txt with filename | description lines, or JSON/YAML? And during development, are placeholder entries fine? Yes 8. Upload persistence. "Saved in a specific folder and added to the Sample List" - should uploads persist across sessions (saved to disk permanently) or only for the current session? permanently 9. Configurator collapse behavior. The sketch shows the configurator can collapse to a small icon. When collapsed, can the user still swap images, or must they expand it first? They must expand it first.

C. NMS Section 10. Step granularity. What constitutes "one step" in the timeline? My assumption: one step = "pick the highest-confidence box, compare it against all remaining boxes, suppress those above IoU threshold." Confirm? One Iteration of the loop 11. Confidence threshold application. Is the confidence threshold applied (a) as a pre-filter before NMS starts (step 0), or (b) continuously? I'd assume (a) - it's the standard pipeline. Yes (a) standard 12. Soft-NMS variant. Soft-NMS has two common forms: linear and Gaussian. Which do you want? Both, with a selector? And what sigma default for Gaussian? Both and leave adjust for sigma 13. Comment section interaction. "Each box in question shall be highlighted in the loaded image if pressed in the comment section." So clicking a log entry like "IoU(box_3, box_7) = 0.62" highlights both boxes 3 and 7? Confirm. Yes 14. Bounding boxes list vs. Keep list updates. When a box is suppressed, does it (a) leave the Bounding boxes list and go nowhere, (b) move to a "suppressed" list, or (c) just get struck through? Sketch is ambiguous. Leaves to suppressed 15. Timeline scrubbing. "User can browse through it" - does scrubbing back re-render the image at that step's state (i.e., the simulator is fully time-reversible)? I'll assume yes. yes

D. AP / mAP Section 16. Greyout trigger. "Greyout until all steps in NMS run." Does this mean greyed out until the user clicks "Run All Steps," or until they've manually stepped through every single one? I'd assume the former. Yes 17. Ground truth requirement. AP/mAP needs ground-truth labels. For sample COCO images, labels are available. For uploaded images without uploaded labels, AP/mAP cannot be computed. How should the UI handle this? Disable the section? Show a warning? Don't allow upload of images without labels 18. AP calculation method. Two standards exist: 11-point interpolation (PASCAL VOC 2007), all-point interpolation (PASCAL VOC 2010+/COCO). The sketch says "number of points for calculation" which suggests N-point interpolation is user-configurable. Confirm? And is there a default? Configurable. Default 11. 19. mAP IoU threshold. Is mAP computed at a single IoU (e.g., 0.5) or COCO-style averaged over [0.5:0.05:0.95]? Configurable? Configurable. 20. "Recall" button. The requirements say "There shall be a button to so a recall." I genuinely cannot parse this – is "recall" a typo? Did you mean "recompute" / "rerun"? Or a button that displays a recall-specific metric/curve? – remove shall run automatically. 21. Per-class PR graphs. The sketch shows Class1 / Class2 boxes with what appears to be a PR curve. Should the user click a class to see that class's PR curve, with the main AP/mAP plot reflecting the selection? yes

E. Sketch-specific items I want to confirm I'm reading correctly 22. Top-right "4°" / "1°" markings in the sketch – are those step indicators, or just numbering on your sketch? I'm assuming the latter (page annotations). 23. The dashed list under the configurator showing Image1/2/3 – this appears when "Sample" or "Examples" is pressed, correct? And selecting one populates "Loaded Image"? 24. Two boxes inside "Loaded Image" with class: 0.2 and class: 0.8 – these are example bounding boxes labeled with confidence, right? Not actual class IDs. yes 25. The triangular "play" and "fast-forward" icons next to the timeline – single-step and run-all-steps respectively? yes 26. The PR curve at bottom-left appears to be the main AP/mAP visualization, with Precision on Y and Recall on X. Confirm orientation (it's a bit rotated in the photo). yes 27. The table at the bottom with "Conf. Threshold | Recall | Precision" columns – is this a per-threshold breakdown table feeding the PR curve? How many rows / which thresholds? Just as interim steps

F. Things entirely missing that I'd want to lock down 28. Color scheme / theme. Light, dark, either? – LIght 29. Error states. What happens if YOLO returns zero detections? If the IoU threshold is set to 0 or 1? Give warning 30. Persistence. Should simulator state survive a page refresh, or always start fresh? Start fresh 31. Browser support. Any minimum (Chrome only, or all modern)? modern 32. Language/i18n. English only, I assume?yes